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Broadband sensitivity improvement via coherent quantum feedback with PT symmetry

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A conventional resonant detector is often subject to a trade-off between bandwidth and peak sensitivity that can be traced back to quantum Cramer-Rao Bound. Anomalous dispersion has been shown to improve it by signal amplification and is thus more robust against decoherence, while it leads to instabilities. We propose a stable quantum amplifier applicable to linear systems operating at the fundamental detection limits, enabled by two-mode non-degenerate parametric amplification. At threshold, one mode of the amplifier forms a $\mathcal{PT}$-symmetric system of original detector mode. Sensitivity improvements are shown for laser-interferometric gravitational-wave detectors and microwave cavity axion detectors.

Introduction.— Oscillators are often used to measure weak classical forces. In the early days of gravitational-wave (GW) detection, excitations of mechanical oscillators (resonant bars) were read out by inductive, capacitive [1] or parametric [2] transducers and via a superconducting quantum interference device (SQUID) amplifer [1–3]. A more sensitive technique is to read out the phase fluctuations in a light beam (Michelson and Fabry-Perot type interferometers) [4–6] using photodetection. Signal recycling [7] and Resonant Sideband Extraction (RSE) [8] techniques were designed optimize GW sensitivity by tailoring the optical frequency response. In this process, Mizuno noticed a trade off between bandwidth and peak sensitivity [9] – analogous to the gain-bandwidth product in electronic amplifiers [10]. Braginsky *et al.* [11, 12] showed [1], using the energy-phase uncertainty relation, that the power spectral density of equivalent spacetime strain noise is $S_h(\Omega) \geq 4\hbar^2/S_{\mathcal{E}}(\Omega)$ where $S_{\mathcal{E}}$ is the spectral density of energy in the cavity, and

$$\int_0^{+\infty} d\Omega/(2\pi) S_h^{-1}(\Omega) \leq \Delta\mathcal{E}^2/(4\hbar^2)\,. \tag{1}$$

This integral embodies a *bandwidth-sensitivity trade-off*. This was also obtained by Tsang, et al. using Quantum Fisher Information [13], and further elaborated in Refs. [14–17]. For coherent states, $\Delta\mathcal{E}^2 = \hbar\omega_0\mathcal{E}$, and in this case Eq. (1) is also referred to as the *Energetic Quantum Limit* (EQL) for GW detection [2]. The EQL trade-off applies to all quantum metrology experiments that use oscillators at coherent states.

Detectors.— In this paper, we consider two types of detectors. GW detectors (of the laser interferometer type [6]) use optical resonators to increase the interaction between the spacetime strain and the laser light field. Axion detectors [20] (of the Sikivie Haloscope [21] type) use microwave resonators and a powerful permanent magnet to increase the interaction with the axion field and the microwave field. In both cases, the emphasis is on increasing the quality factor [22], Q, of the resonance so as to maximize the transduction coefficient between the physical signal and an electrical readout variable. However, the EQL limits the useful bandwidth of the detector according to the gain-bandwidth tradeoff in Eq. (1).

In GW detection, the EQL can be surpassed when nonclassical states of light are created for the arm cavity, with $\Delta\mathcal{E}^2 > \mathcal{E}^2/\bar{N}$ [23–26]. This can be implemented via squeezing injection [27], or internal ponderomotive squeezing achieved by optical springs [14, 18, 19]. To directly address the EQL, another concept of White-Light Cavity (WLC), which resonates with a broader spectrum of frequencies without sacrificing sensitivity, has been proposed [23–26, 28–32]. Despite beating the EQL in theory, in some cases [25, 26], improving sensitivity by suppressing noise, while keeping signal at the same level, makes the system more susceptible to optical losses — similar to the situation when injecting squeezed vacuum [27, 33].

In the unstable WLC (uWLC) design of Miao *et al.* [24] (Fig. 1), an additional *coherent quantum feedback controller* [34, 35] (enabled by an *unstable* optomechanical oscillator) is attached to a laser interfereomter to provide an "anomalous dispersion" whose *negative group time delay* cancels the positive group time delay in the interferometer, achieving broadband signal amplification without increasing noise. The instability of this quantum system can, in principle, be stabilized by a classical controller without adding noise [18, 19]

In this paper we introduce a parity-time ($\mathcal{PT}$)-symmetric coherent quantum strategy [36, 37] that not only leads to a stable WLC (sWLC), but also applies to a wide range of quantum systems. We first consider the simplest sWLC, showing that it approaches an *Exceptional Point* (EP) [38, 39] as the feedback gain reaches the threshold of stability, at which point the

[1] More specifically, they assumed a single cavity optimally aligned with an incoming plane GW with wavelength much larger than the cavity length.
[2] Refs. [11–16] all used the double-sided definition for spectral density, while in this paper, in Eq. (1), we have used the single-sided definition in order to be consistent with most GW literature [18, 19].

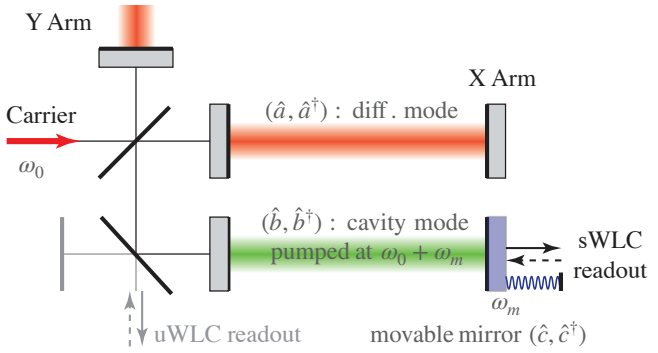


FIG. 1. Coherent quantum feedback for laser interferometer GW detectors. In the uWLC scheme (signal readout shown in gray), positive storage time in the arms is canceled by anomalous dispersion of the filter cavity, which has a movable mirror with mechanical oscillation frequency ω_m and is pumped at $\omega_0 + \omega_m$. The *controller* is the time reversal of the *plant*. The sWLC scheme extracts signal at the end of the filter cavity (shown in black). Both schemes are also illustrated in Fig. 2.

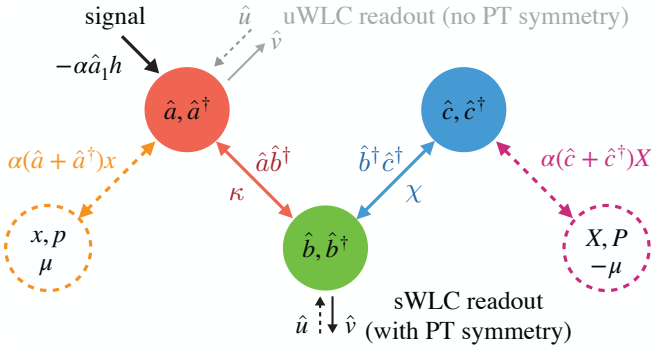


FIG. 2. A force sensor [mode $(\hat{a}, \hat{a}^\dagger)$, plus (x, p) for the test mass in case of GW detectors] plus coherent quantum feedback with modes $(\hat{b}, \hat{b}^\dagger)$, $(\hat{c}, \hat{c}^\dagger)$ and (X, P). Instead of directly coupling to readout, the mode $\hat{a}$ first couples to a mode $\hat{b}$ (via $\hat{a}^\dagger\hat{b}$), which in turn couples to $\hat{c}$ (via $\hat{b}^\dagger\hat{c}^\dagger$). Mode $\hat{c}$ may also couple to (X, P) with negative mass to cancel the back-action noise caused by $\hat{a}$'s coupling to (x, p) (back-action evasion, see Fig. 5). The mode $\hat{b}$ is coupled to an external continuum ("sWLC readout" in black). System is $\mathcal{PT}$ symmetric when $\kappa = \chi$. The uWLC scheme is recovered when $\hat{a}$ couples to continuum instead (shown in gray); further setting $\kappa \to 0$ reduces to a conventional detector.

theoretical gain in sensitivity is infinite. We then show that the design strategy is very general, since the controller's Hamiltonian always corresponds to the time reversal of that of the plant. We finally discuss applications to GW and axion detection.

Stable Optomechanical Amplifer.— Suppose the mode of a weak-force sensor has annihilation/creation operators $(\hat{a}, \hat{a}^\dagger)$, which can be reorganized into quadrature operators $\hat{a}_1 = (\hat{a} + \hat{a}^\dagger)/\sqrt{2}$, $\hat{a}_2 = (\hat{a} - \hat{a}^\dagger)/(\sqrt{2}i)$, and the detector is coupled to a signal h via $\hat{V}_s = -\alpha\hat{a}_1 h$. For a GW detector, $\hat{a}$ is the cavity mode, h is GW strain, while α is achieved by strong carrier field in the cavity (we first ignore radiation-pressure

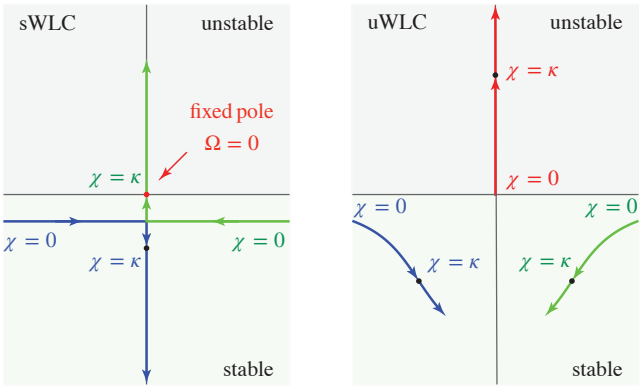


FIG. 3. Trajectory of poles of WLC dynamics in the complex plane (dashed curves), with direction of increasing χ (with κ fixed) indicated by arrows. Left: the sWLC has a fixed pole at $\Omega = 0$ (independent from χ), which is invisible to the input-output relation (4); the other two poles first move toward the imaginary axis and meet each other, then move up and down separately, until one of them reaches 0 when $\chi = \kappa$. For $\chi > \kappa$, the system becomes unstable. Right: on contrary, the uWLC is unstable as soon as $\chi > 0$.

effects [27]; h will be replaced by the axion field Ψ later in the paper).

As in Fig. 2, let us couple $\hat{a}$ to $\hat{b}$ with rate κ, while $\hat{b}$ and $\hat{c}$ are together amplified by a non-degenerate amplifier with rate χ, the Hamiltonian reads (Sec. 2 of Ref. [40]):

$$\hat{V} = i\hbar\kappa(\hat{a}\hat{b}^\dagger - \hat{a}^\dagger\hat{b}) + i\hbar\chi(\hat{b}^\dagger\hat{c}^\dagger - \hat{b}\hat{c}). \tag{2}$$

The mode $\hat{b}$ is read out by being coupled to a continuum $\hat{w}_\xi$ of incoming ($\xi < 0$) and out-going ($\xi > 0$) fields at $\xi = 0$, with $\hat{V}_b = i\sqrt{2\gamma_R}\hat{b}\hat{w}^\dagger_{\xi=0} + \int_{-\infty}^{+\infty} d\xi \left[i\hat{w}^\dagger_\xi \partial_\xi \hat{w}_\xi + \text{h.c.}\right]$, and we shall use $\hat{u}$ and $\hat{v}$ to denote $\hat{w}_{0-}$ and $\hat{w}_{0+}$, the field right before entering and right after leaving. (In uWLC the operator $\hat{a}$, instead of $\hat{b}$, is coupled to the continuum $\hat{w}_\xi$.) The resulting Heisenberg equations are given by

$$\dot{\hat{a}} = i\alpha h - \kappa\hat{b},\ \dot{\hat{c}}^\dagger = \chi\hat{b},\ \dot{\hat{b}} = -\gamma_R\hat{b} + \kappa\hat{a} + \chi\hat{c}^\dagger + \sqrt{2\gamma_R}\hat{u}, \tag{3}$$

and $\hat{v} = \hat{u} - \sqrt{2\gamma_R}\hat{b}$. The phase quadrature of the output field reads:

$$\hat{v}_2 = \frac{\Omega^2 - i\gamma_R\Omega + \chi^2 - \kappa^2}{\Omega^2 + i\gamma_R\Omega + \chi^2 - \kappa^2}\hat{u}_2 + \frac{2\sqrt{\gamma_R}\kappa\alpha}{\Omega^2 + i\gamma_R\Omega + \chi^2 - \kappa^2}h\,. \tag{4}$$

From poles of $\hat{v}_2$, we see that increasing the amplification rate χ tends to destabilize the system, while increasing κ tends to stabilize it; the entire system is stable when $\chi \le \kappa$. The system also has a fixed pole at $\Omega = 0$ not visible in $\hat{v}_2$. We plot trajectories of poles for sWLC and uWLC dynamics in Fig. 3.

In the single-cavity (or conventional) detector (i.e., $\hat{a}$ is read out directly, without adding $\hat{b}$ and $\hat{c}$) has: $\hat{v}_2 = (\Omega - i\gamma_R)/(\Omega + i\gamma_R)\hat{u}_2 - 2i\sqrt{\gamma_R}\alpha h/(\Omega + i\gamma_R)$. For vacuum fluctuations, we have the spectrum $S_{\hat{u}_i\hat{u}_j} = \delta_{ij}$, and thus

$$S_h^{\text{con}} = \frac{\Omega^2 + \gamma_R^2}{2\gamma_R\alpha^2},\ S_h^{\text{amp}} = \frac{(\Omega^2 - \kappa^2 + \chi^2)^2 + \gamma_R^2\Omega^2}{2\gamma_R\kappa^2\alpha^2}. \tag{5}$$

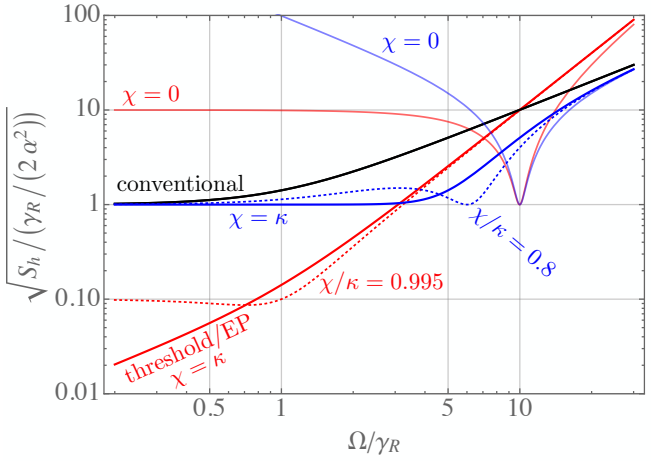


FIG. 4. Coherent amplification of sensitivity by sWLC (red) and uWLC (blue) when $\kappa = 10\gamma_R$ for a lossless single-mode detector; Connection between threshold and EP can be found in the main text. For the same choice of γ_R and χ, uWLC tends to improve higher frequency sensitivity, while sWLC tends to improve lower frequency sensitivity.

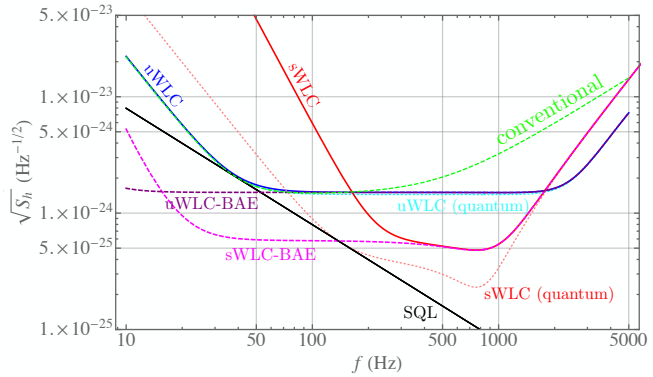


FIG. 5. Example of GW noise spectra for sWLC and uWLC (both with $\kappa/(2\pi) = 5\,\mathrm{kHz}$, $\chi/(2\pi) = 4.93\,\mathrm{kHz}$) configurations, *only* accounting for quantum noise and thermal noise from the mechanical oscillator $\hat{c}$. Solid red and blue curves are sWLC and uWLC (with dotted pink and cyan curves showing the quantum noise contribution), while dashed magenta and purple curves are BAE configurations that require additional nonlinear elements. Shown for comparison is the quantum noise spectrum for the conventional single cavity (using tuned RSE configuration) without optical loss. Note that in this figure, sWLC, uWLC and conventional are plotted with the same carrier wavelength, arm cavity power, $\hat{c}$-mode mechanical loss (only for WLC), thermal bath condition (environment temperature), and the readout rate $\gamma_R/(2\pi) = 500\,\mathrm{Hz}$.

A conventional detector has $\int_0^{+\infty} d\Omega/(2\pi)[S_h^{\rm con}]^{-1} = \alpha^2/2$ and the gain in integrated sensitivity by amplification is

$$\Lambda \equiv \frac{\int_0^{+\infty} d\Omega/(2\pi)[S_h^{\rm amp}]^{-1}}{\int_0^{+\infty} d\Omega/(2\pi)[S_h^{\rm conv}]^{-1}} = (1-\chi^2/\kappa^2)^{-1} \tag{6}$$

with $\Lambda \to +\infty$ as $\chi \to \kappa-$. Both configurations have the same noise level in $\hat{v}_2$, while the amplifier has higher signal amplitude, especially as $\chi \to \kappa-$. The broadband improvement for the case $g = 10\gamma$ is shown in Fig. 4.

Effects of Decoherence.— Suppose the $(\hat{a}, \hat{b}, \hat{c})$ modes have dissipation rates γ_a, γ_b and γ_c, respectively, to their baths, and have noises $\hat{n}_a^L$, $\hat{n}_b^L$, $\hat{n}_c^L$ entering, we need to apply

$$\dot{\hat{Y}} = ... \to \dot{\hat{Y}} = ... - \gamma_Y \hat{Y} + \sqrt{2\gamma_Y}\hat{n}_Y^L, \tag{7}$$

with $Y = a, b, c$ to Eq. (3). Depending on the different nature of noise sources, the decay rate γ_Y and noise spectrum S_Y^L vary. [3]

Connection to $\mathcal{PT}$ symmetry and generalization to multi-mode systems.— On threshold at $\chi = \kappa$, poles of the output field $\hat{v}_2$ are at $\Omega = 0, -i\gamma_R$, making $\Omega = 0$ a double pole (see Fig. 3). At this point, the system only has two independent eigenmodes, making it an Exceptional Point (EP) [38, 39]. The system is also $\mathcal{PT}$ symmetric: the Hamiltonian is invariant if exchanging $\hat{a} \to \hat{c}^\dagger$ and $\hat{c} \to \hat{a}^\dagger$. Specifically, the $\mathcal{P}$ (parity) transformation swaps the $(\hat{a}, \hat{a}^\dagger)$-mode and the $(\hat{c}, \hat{c}^\dagger)$-mode, the $\mathcal{T}$ (time-reversal) transformation swaps creation and annihilation operators; combination of $\mathcal{PT}$ transformation leaves $\hat{b}$ invariant.

[3] See Supplemental Material [] for more details on decoherence.

Now, for a multi-mode device with modes $\{\hat{a}_j\}$ coupled to signal h with coefficients $\{\alpha_j\}$ and with $\sum \beta_j \hat{a}_j$ originally coupled to the external continuum, let us bring in systems $\{\hat{c}_j\}$, with $\{\hat{c}_j^\dagger\}$ having the same equations of motion as $\{\hat{a}_j\}$, and the mode $\hat{b}$, which couple to $\{\hat{a}_j\}$ and $\{\hat{c}_j^\dagger\}$:

$$\dot{\hat{a}}_j = \mathcal{M}_{jk}\hat{a}_k + i\alpha_j h - \kappa\beta_j\hat{b}, \quad \dot{\hat{c}}_j^\dagger = \mathcal{M}_{jk}\hat{c}_k^\dagger + \chi\beta_j\hat{b}, \tag{8}$$

$$\dot{\hat{b}} = -\gamma_R\hat{b} + \kappa\beta_j\hat{a}_j + \chi\beta_j\hat{c}_j^\dagger + \sqrt{2\gamma_R}\hat{u}\,, \quad \hat{v} = \hat{u} - \sqrt{2\gamma_R}\hat{b}. \tag{9}$$

These equations can be solved in a matrix form and lead to:

$$\hat{v} = \left(\frac{\Omega - i\gamma_R - i\Delta}{\Omega + i\gamma_R - i\Delta}\right)\hat{u} - i\sqrt{2\gamma_R\kappa}\left(\frac{\sum_{k,l}\beta_k G_{kl}\alpha_l}{-i\Omega + \gamma_R - \Delta}\right)h, \tag{10}$$

where $\Delta = (\chi^2 - \kappa^2)\sum_{k,l}\beta_k G_{kl}\beta_l$, $\mathbf{G} = (-i\Omega\mathbf{I} - \mathcal{M})^{-1}$. Similar to the single-mode case, $\mathcal{PT}$-symmetry at $\chi = \kappa$ allows $\Delta = 0$, hence the cancellation of inertia existed for $\{\hat{a}_j\}$, and leads to a broadband amplification of signal: noise remains white while signal transfer function is proportional to κ.

Application to laser interferometer GW detectors – The sWLC can be implemented in a GW detector by making a simple modification of the uWLC: changing the location of signal extraction, as shown in Fig. 1. To incorporate motions of mirrors under radiation pressure, we need to modify the interaction Hamiltonian to include $\hat{x}$ and $\hat{p}$ of the differential mode of mirror motion [27]:

$$\hat{V}_{\rm GW} = -\alpha_{\rm GW}(\hat{x} - Lh)\hat{a}_1 + \hat{p}^2/(2\mu) \tag{11}$$

Here $\alpha_{\rm GW} = \sqrt{2P_c\hbar\omega_0/(Lc)} = \sqrt{\mathcal{E}\hbar\omega_0}$, with P_c the circulating power in the arm ($\mathcal{E}$ is the energy stored in each

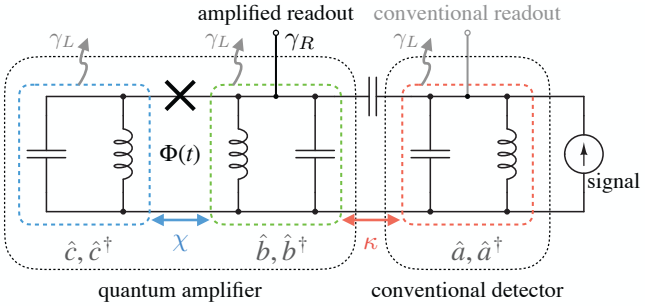


FIG. 6. Circuit diagram of an axion detector (dashed box on the right) attached to a quantum amplifier (dashed box on the left). The parametric amplifier is achieved by Josephson 3-wave mixer indicated by the × symbol in the figure, with $\Phi(t)$ indicating time-depending external flux that act as parametric pumping (see Fig. 9 of Ref. [47]). Coupling "amplified readout" to a continuum results in a sWLC, while keeping "conventional readout" coupling to a continuum will result in uWLC.

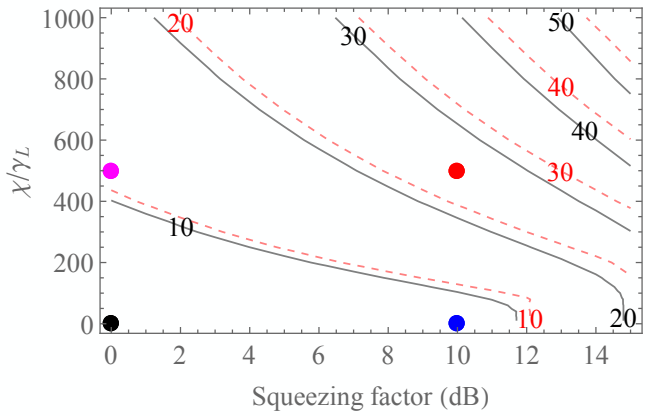


FIG. 7. The enhancement of the scan rate $\mathcal{R}_a$ (12) achievable by sWLC axion detectors over the single-cavity one, as a function of the amplifier gain χ and squeeze factor e^{-2r}, optimized over the rate of $\hat{a}$-$\hat{b}$-mode coupling κ and readout coupling γ_R. The uWLC cases optimized with the same procedure are indicated by dashed red lines for comparison, showing comparable but slightly less improvements over the single-cavity configuration. The single-cavity and sWLC configurations shown in Fig. 8 are indicated by the correspondingly colored dots.

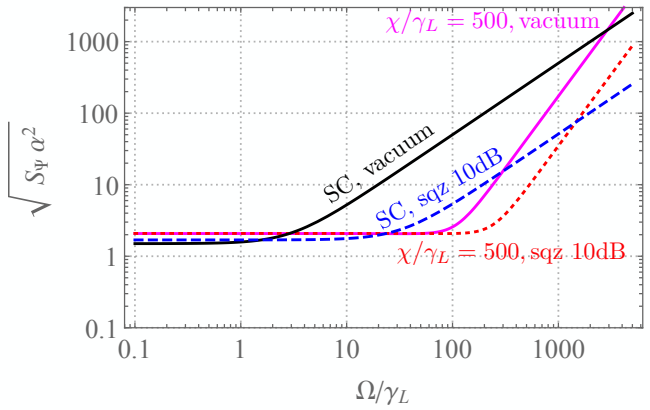


FIG. 8. Examples of single-cavity (SC) and sWLC noise spectrum with $\mathcal{R}_a$-optimized κ, γ_R for different χ and input squeezing levels (sqz) e^{-2r} shown by color dots in Fig. 7. See the main text for more details.

arm), ω_0 is the carrier angular frequency, L is the arm length, and $\mu = M/4$ is the reduced mass of the differential mode (M is the mass of each mirror). In Fig. 5, we plot quantum noise spectrum (including radiation-pressure noise) for sWLC and uWLC configurations with $\gamma_R/(2\pi) = 500\,\text{Hz}$, $(\kappa,\chi)/\gamma_R = (10, 9.86)$. We use the LIGO Voyager [41] parameters $M = 200\,\text{kg}$, $L = 4\,\text{km}$, circulating power of 3 MW and a $2\,\mu\text{m}$ laser wavelength. We did not include the effect of optical losses, but did consider thermal noise of the $\hat{c}$ oscillator mode [Eq. (7)], with $Q_m \sim 8\times10^9$ and $T = 4\,\text{K}$ [24, 42], which is the most serious source of decoherence in our scheme.

Our proposal may also be realizable by inserting a nonlinear crystal into the signal recycling cavity, and pump with the sum frequency of the carrier frequency and another mode of the SR cavity; this can also be viewed as a non-degenerate quantum expander [26, 43]. Furthermore, if we demand a full $\mathcal{PT}$ symmetry to include the movable mirrors, we need to couple to an additional mode $(\hat{X}, \hat{P})$ that has a *negative effective mass* with Hamiltonian $\hat{V}_{\text{BAE}} = -\alpha_{\text{GW}}\hat{X}\hat{c}_1 - \hat{P}^2/(2\mu)$ analogous to the proposal of Ref. [44]. We shall leave the details of this for future work, but do show their potentially achievable sensitivity in Fig. 5, in magenta and purple curves; aside from quantum noise, we have only included $\hat{c}$-oscillator thermal noise. This significant potential for improvement motivates further studies into this direction [45, 46].

Application to microwave axion detectors.– For an axion detector consisting of a stationary magnetic field $\mathbf{B}_0$ and a single mode of a microwave resonator, the interaction Hamiltonian can be written as $\hat{V}_{\text{axion}} = \alpha_{\text{axion}}(\Psi_1\hat{a}_1 + \Psi_2\hat{a}_2)$ where $\hat{a}_{1,2}$ are the mode quadratures, and $\Psi_{1,2}$ are two quadratures of the oscillations of the axion field $\mathcal{A}(t) = \Psi(t)e^{-i\omega_0 t} + \Psi^*(t)e^{i\omega_0 t}$ with $\Psi_1 = (\Psi + \Psi^*)/\sqrt{2}$, $\Psi_2 = (\Psi - \Psi^*)/(\sqrt{2}i)$. Here ω_0 is the central frequency at which the device operates; it may need to be scanned over different values to search for the yet unknown Compton frequency of axion.

A proposed WLC circuit that realize the Hamiltonian (2) is shown in Fig. 6. The parametric amplification can be realized by a 3-wave mixing device [47, 48]; sWLC corresponds to having $\hat{b}$ mode coupled to the external continuum ("amplified readout" in the figure), while uWLC corresponds to having $\hat{a}$ mode coupled to the external continuum ("conventional readout" in the figure). We will consider one of the axion-field quadrature Ψ_1, which is read out from v_2, and account for squeezing injection by $S_{\hat{u}_2\hat{u}_2} = e^{-2r}$, where $S_{\hat{u}_2\hat{u}_2}$ is the spectral density of the phase quadrature of the input field, as in Eq. (5). We also assume the same loss rate γ_L for each of the $\hat{a}$, $\hat{b}$ and $\hat{c}$ modes, assuming vacuum noise from each bath, and propagate these noise contributions to the output by applying

Eq. (7) to Eq. (3) (h should now be replaced by Ψ_1), leading to Ψ_1-referred noise spectrum S_Ψ.

As a figure of merit for axion detection, we shall use the effective scan rate [49, 50]:

$$\mathcal{R}_a \equiv \int_0^{+\infty} d\Omega/(2\pi) S_\Psi^{-2}(\Omega), \tag{12}$$

which is inversely proportional to the search time it takes to detect the axion field, assuming the field's coherence time to be very short compared with observation time, *and* assuming no additional time cost when switching between configurations centered at different frequencies. In comparison with the EQL (1), $\mathcal{R}_a$ (12) favors peak sensitivity more than bandwidth. Considering the losses, numerical optimization is carried out on the scan rate $\mathcal{R}_a$ enhancement: for values of χ/γ_L and e^{-2r}, search over κ/γ_L and γ_R/γ_L to achieve an optimal $\mathcal{R}_a$, and compare it with that of a single-cavity detector with the same γ_L (for which the optimal γ_R/γ_L is 2 [49]). As shown in Fig. 7, the amplifier gain χ improves $\mathcal{R}_a$ substantially, and in a way compatible with squeezing. In Fig. 8, we plot the noise spectra of several configurations. These configurations improve the scan rate $\mathcal{R}_a$ (12) by substantially broaden the detector bandwidth, thereby requiring *far* less switches between different central frequencies.

Conclusions.— We have proposed a quantum amplifier that can be attached to an existing sensor as a coherent quantum feedback device, and improve its sensitivity-bandwidth product. The improvement is achieved via signal amplification, and therefore, is robust against readout losses. Our sensitivity gain can be connected to $\mathcal{PT}$ symmetry: a mode $\hat{b}$ is coupled to our original sensor modes $\{\hat{a}_j\}$, as well as auxiliary modes $\{\hat{c}_j\}$, which is time reversal to the original sensor modes. Such a strategy can be used to substantially improve the sensitivity of laser interferometer gravitational-wave detectors, and microwave axion detectors. We analyzed different figures of merit (Eqs. (6) & (12)) for the two detector schemes, while a more complete PT-symmetric structure for backaction evasion requires further study. In previous studies of $\mathcal{PT}$ symmetry, the balance of gains and losses are often emphasized, usually in the context of non-Hermitian Hamiltonians that are constructed by eliminating external degrees of freedom [51, 52]. In our case, the $(\{\hat{a}_j\}, \hat{b})$ subsystem has loss, the $(\{\hat{c}_j\}, \hat{b})$ subsystem has (the balancing) gain, and the entire system has a Hermitian Hamiltonian. In this way, we are able to arrive at features similar to previous work without having to resort to external degrees of freedom.

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Note added.–During the completion of this work, we became aware of a related but different study [53].

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